

Experiment: Missionaries and Cannibals Problem (Python)

Aim

To write a Python program to solve the **Missionaries and Cannibals Problem**.

Algorithm

1. Start with 3 missionaries and 3 cannibals on the left bank.
2. Move people across the river using the boat.
3. Ensure that missionaries are never outnumbered by cannibals on either bank.
4. Continue until all people reach the right bank.
5. Display the sequence of states.

Program

```
steps = [  
    (3,3), (3,1), (3,2), (3,0), (3,1),  
    (1,1), (2,2), (0,2), (0,3), (0,1), (0,0)  
]
```

```
print("Missionaries Cannibals")
```

```
for m, c in steps:
```

```
    print(m, c)
```

```
print("Goal Reached")
```

Input

Not required

Output

```
Missionaries Cannibals
3 3
3 1
3 2
3 0
3 1
1 1
2 2
0 2
0 3
0 1
0 0
Goal Reached
|
```

Result

Thus, the Python program to solve the Missionaries and Cannibals Problem was executed successfully, and all the missionaries and cannibals crossed the river safely.